

# Network Protocols

*-Dr. Sumana Maiti*

# Broadcasting

255.255.255.255 → Limited broadcast address

FF:FF:FF:FF:FF:FF → Broadcast MAC address

AL m

TL 

|   |   |   |
|---|---|---|
| m | x | y |
|---|---|---|

NL 

|   |   |   |                |                 |
|---|---|---|----------------|-----------------|
| m | x | y | I <sub>A</sub> | 255.255.255.255 |
|---|---|---|----------------|-----------------|

DLL 

|   |   |   |                |                 |                |                   |
|---|---|---|----------------|-----------------|----------------|-------------------|
| m | x | y | I <sub>A</sub> | 255.255.255.255 | M <sub>A</sub> | FF:FF:FF:FF:FF:FF |
|---|---|---|----------------|-----------------|----------------|-------------------|

PL 

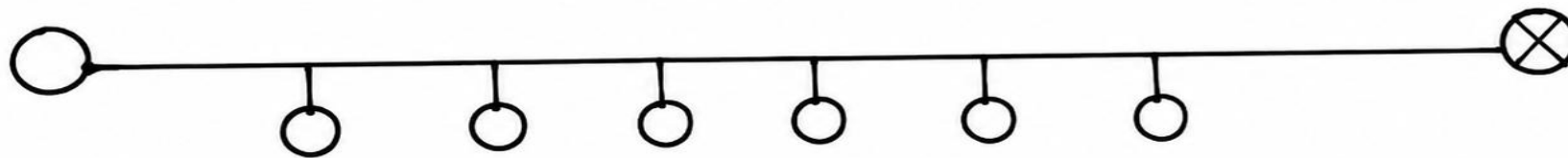
|  |
|--|
|  |
|--|

Destination MAC address

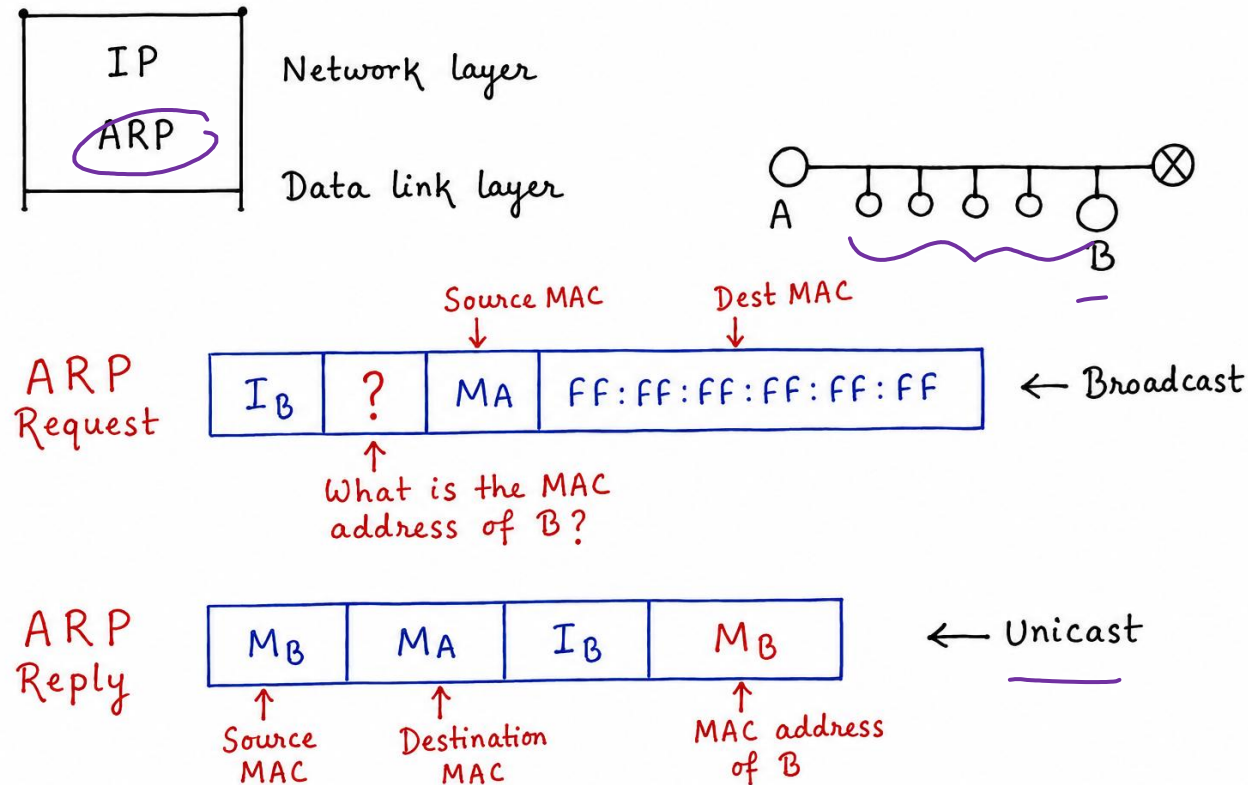
NL

DLL

PL

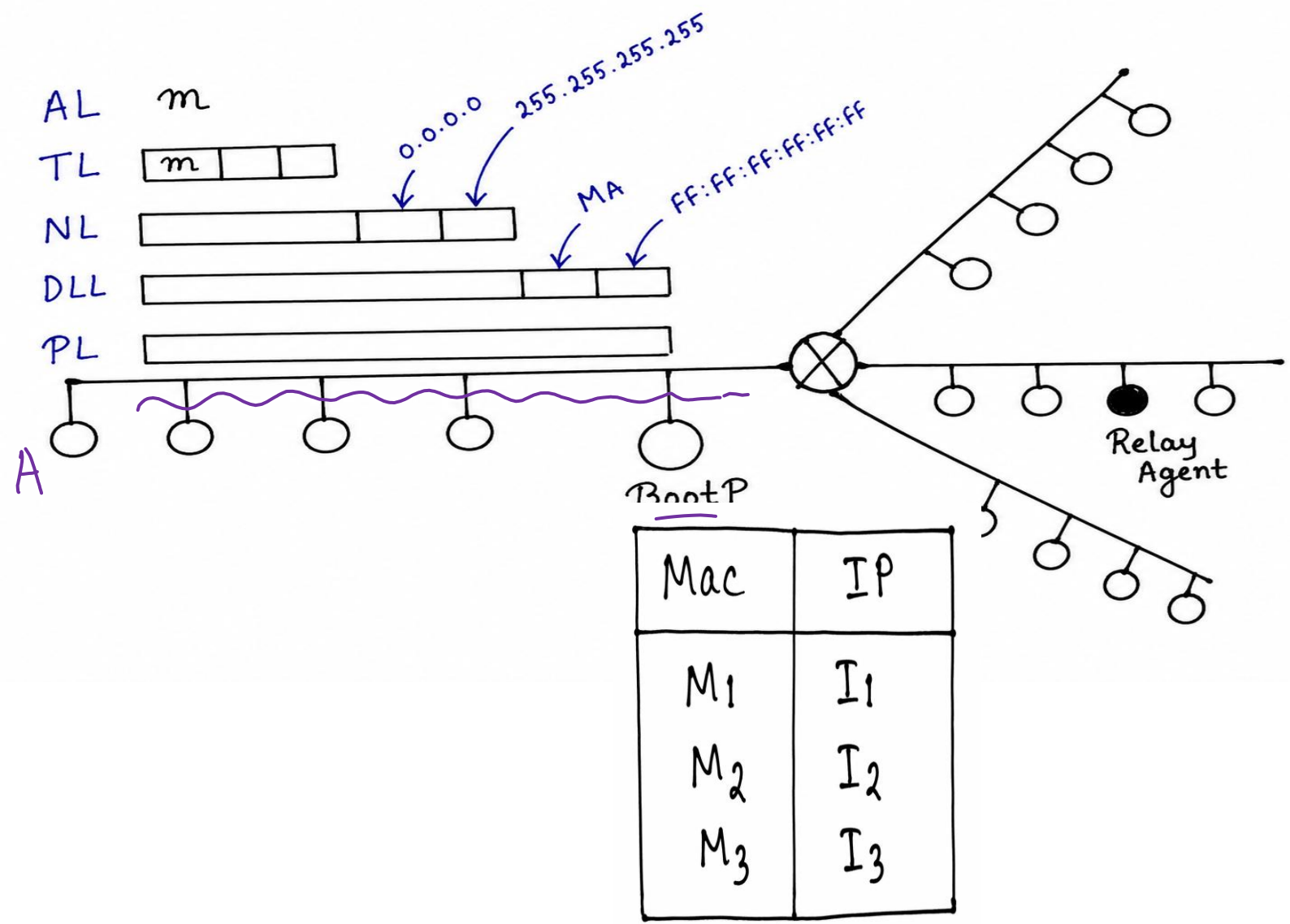


# Address Resolution Protocol (ARP)



- ARP converts IP address to MAC address.
- An IP packet will come to ARP and from ARP, it will go to Data link layer
- ARP request is multicast, ARP reply is ~~broadcast~~ *unicast*

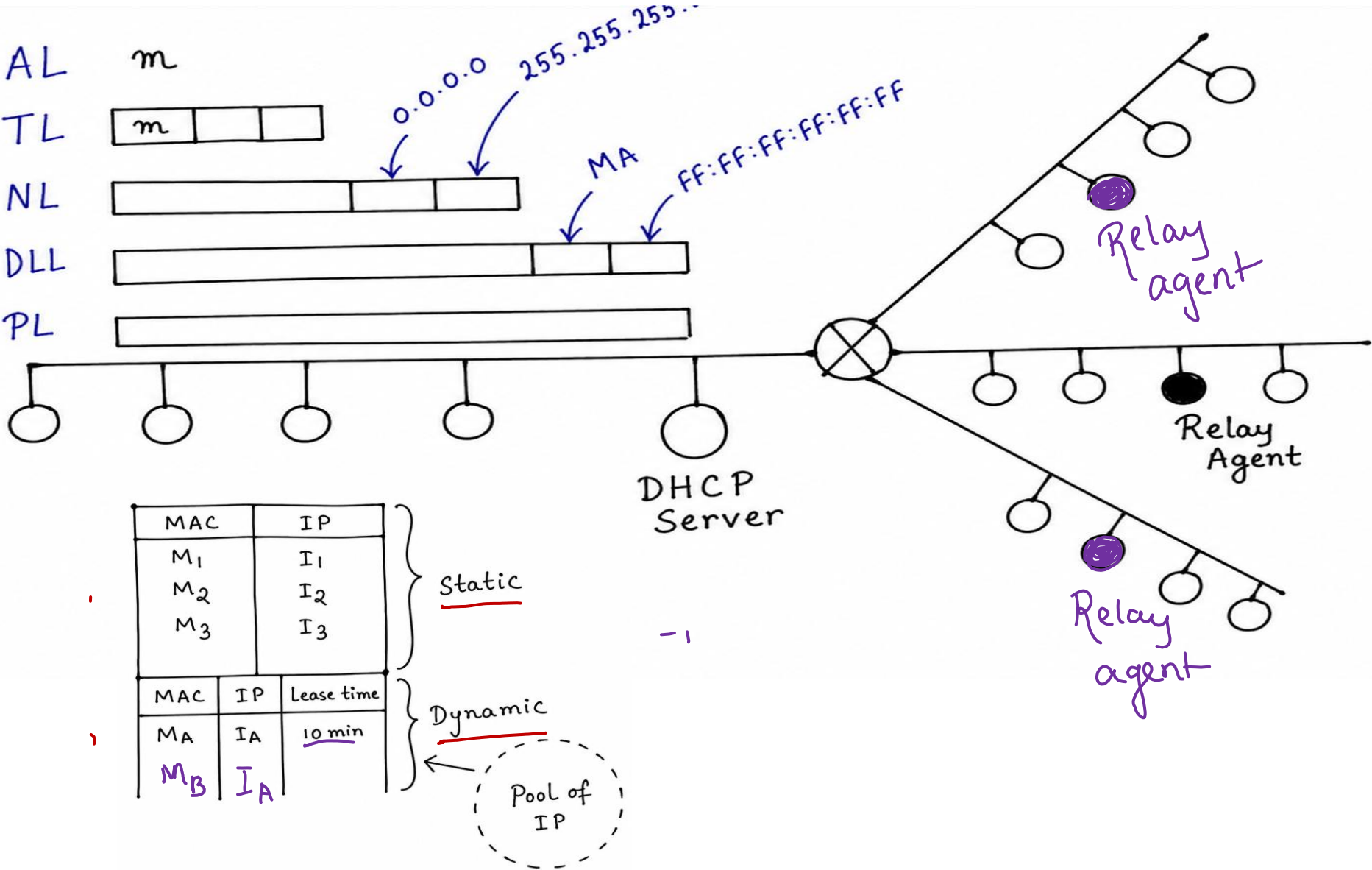
# Bootp



# Bootp

- Bootp converts MAC address to IP address.
- Bootp runs at application layer.
- Request is broadcast and reply is unicast.
- if network does not have a Bootp server, it should have a relay agent.
- Relay agent reads the packet and sends it to Bootp server as a unicast packet. Relay agent knows the IP of Bootp server.
- Bootp server sends the IP to relay agent via router as a unicast packet.
- Relay agent sends unicast reply to host.
- Therefore, in one network we have Bootp server, in other networks, we have relay agents.

# Dynamic Host Configuration Protocol (DHCP)



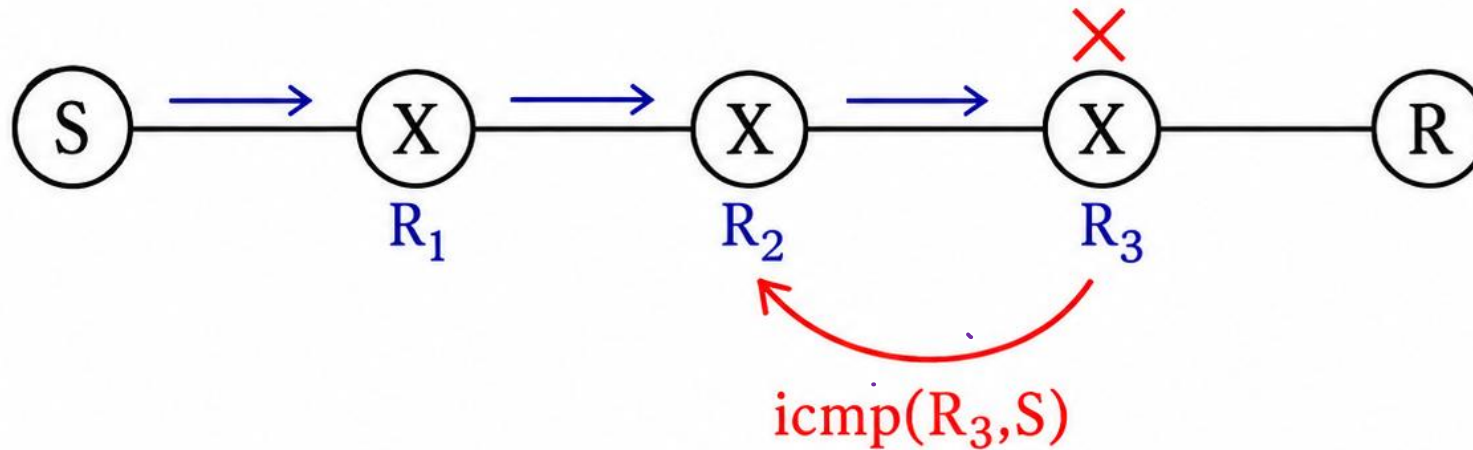
# DHCP

- Dynamic Host Configuration Protocol.
- In DHCP, the table is not static.
- DHCP table has two parts- (a) Static, (b) Dynamic.
- If the host has to be online all the time, static (Permanent IP) is given.
- All the available IP addresses are present in a pool. If user asks IP address, DHCP server picks the IP address from the pool and assign the IP address for a given time. If for more time, the user needs the IP address, the user has to renew the request .

# Internet Control Message Protocol (ICMP)

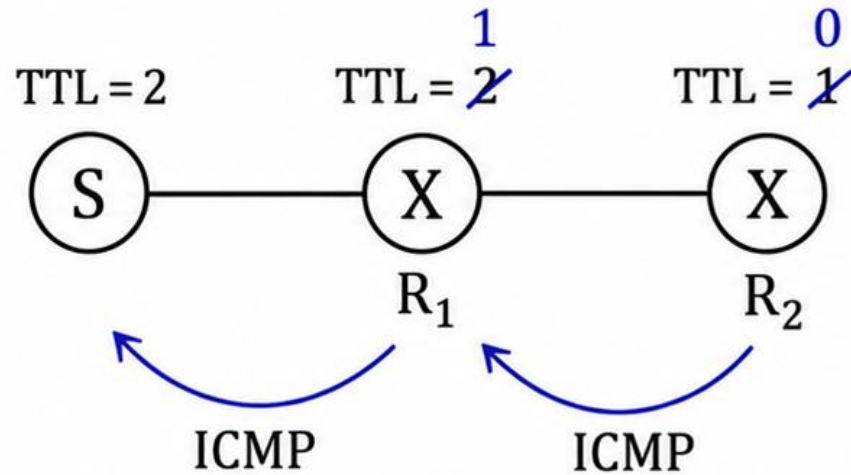
- **Error handling:** Whenever something is wrong, ICMP sends error message. ICMP is assistant of IP packet. If any IP packet is lost, ICMP sends some response
- **Feedback messaging:** TTL exceed, Source quence, Parameter problem, Destination unreachable.
- **Request and Reply:** Checks whether the network layer of the destination and network layers of all the other devices are working or not

# Error Handling



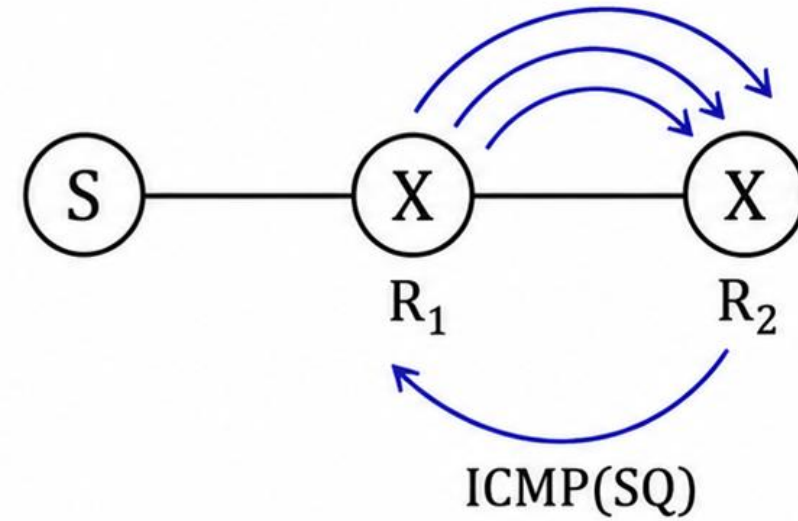
- Anyone who discard the IP packet should send the ICMP packet to Source
- Whenever IP packet is discarded, ICMP packet is generated. But if ICMP packet is discarded, no ICMP packet should be generated.

## TTL exceed



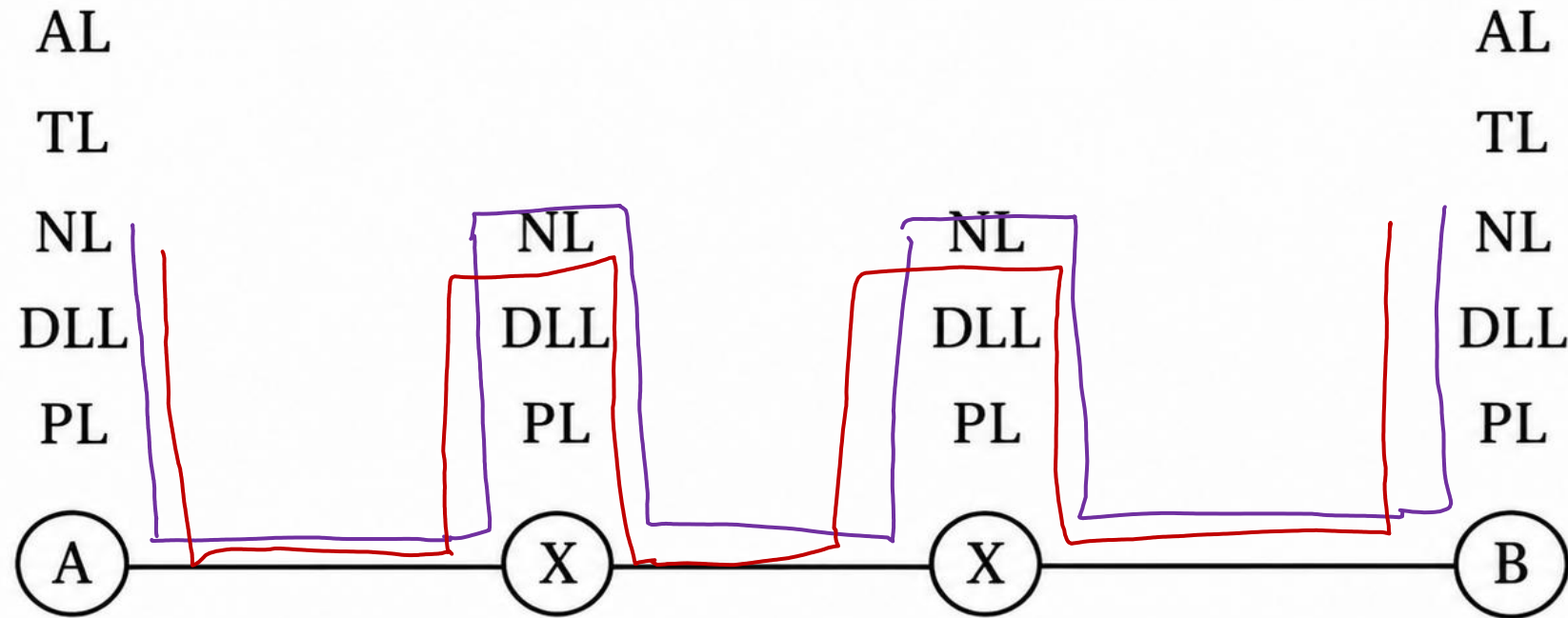
Whenever TTL hits 0, and the packet is not reached the destination, the station is supposed to discard IP packet and send ICMP to Source

## Source quench



If R<sub>1</sub> sends lot of packets to R<sub>2</sub> and R<sub>2</sub> is not able to handle this, R<sub>2</sub> sends ICMP to R<sub>1</sub>

# ICMP request and reply message



Echo request and reply is used to check whether the network layer of the destination is working or not.

"Ping" is ICMP echo request/reply